# The Coach Locus is Offensive and Two-Tiered:
## A Multi-Resolution Event-Study Decomposition of NFL Coaching Effects with a Bounded Defensive Null

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May 2026

**Abstract**

NFL coaching effects are asymmetric across roles, sides of the ball, and observational resolutions. We apply a four-layer decomposition (season-aggregate, within-season event-study, within-game schematic, within-game stress-adaptation) to nflverse play-by-play 2002–2024 across 213 head-coach, 159 OC, and 144 DC off-season changes plus 45 mid-season HC firings. The same partition — the (HC, OC, DC) staff triple at each team-season — is tested at every resolution under identical SNR discipline ($B = 1000$ permutation null, cross-season holdout, propensity-matched event design). The offensive coaching latent is two-locus. Head coaches author macro philosophy: pass-rate ratio $1.36\times$ at mid-season firings ($n = 45, p = 0.002$), pace ratio $1.17\times$ at off-season changes ($n = 142, p = 0.021$). Offensive coordinators author red-zone schematic fine-grain: shotgun-rate ratio $1.34\times$ at off-season OC changes ($n = 175, p = 0.0002$). Naive pooled effects attenuate 30–50% under propensity matching; the conditional estimator that survives is the predictive-validation slope of observed shifts on shifts predicted from each coach's prior-tenure profile. The OC slope is $0.45$ ($n = 151$, 100th percentile of a continuous-staff placebo distribution with median $0.14$), survives leave-one-season-out recalibration within 4%, and is the operational signal used in production projections. The HC slope ($0.31$) is contaminated by team-mean autocorrelation (13th percentile of its own placebo) and is reported at the existence level only (Pearson $r = 0.39, p = 0.004$). Defensive coaching loads chronically, not acutely. At the change-event resolution, twelve defensive cells yield no shift detectable at 80% power against effects $\geq 1.22\times$, while observed offensive effects are $1.34$–$1.36\times$. At the chronic tenure resolution, DC identity is significant on 3 of 7 defensive cells under permutation-null correction; the joint (HC, DC) partition recovers 5 of 7, all on macro tendencies (NET $R^2 0.09$–$0.18$). The asymmetry sharpens into a portability hierarchy: low-level scheme components in fixed-field-position contexts transfer across rosters at a calibrated slope; high-level philosophy is roster-conditional and detectable only as existence.

# 1 Introduction

The convention in NFL projection is to treat coaching as a fixed team effect. This is right for season-aggregate outcomes — pass yards, scoring rate, EPA per play — where roster continuity and franchise context dominate variation. It is not appropriate for schematic dimensions of team identity: formation tendency, tempo, target depth, red-zone play-call mix. The coaching staff is the proximate author of those dimensions. The mismatch produces a small but consistent gap in projection accuracy at exactly the moments when a coaching change occurs. It also produces a larger interpretive problem: a single number for "coaching effect" conflates three things that should be measured separately.

We argue that *does coaching matter* is the wrong question. It collapses three dimensions that single-number tests cannot separate: **resolution**, role, and outcome. A latent that loads cleanly at one resolution can vanish at another for the same data and the same partition. The four-layer decomposition holds the partition fixed and varies only the resolution. Cross-layer differences are therefore attributable to resolution rather than method drift. Single-resolution tests cannot distinguish a real coaching effect that loads at one resolution from a real effect that loads at a different resolution. They cannot distinguish either from a genuine absence.

This paper applies a four-layer decomposition to NFL play-by-play data 2002-2024, using the same partition (coaching staff identity) at four resolutions: season aggregate, within-season event-study around coaching changes, within-game schematic identity by score bucket, and within-game stress adaptation. Every layer uses the same SNR discipline:

permutation-null calibration, cross-season holdout requirement, and propensity-matched event design. The discipline is applied without modification across resolutions.

The main findings are two-fold.

First, the offensive coaching latent is two-locus and hierarchical. The load-bearing finding is a portability slope rather than a pooled mean shift. Conditioning on each coach's prior-tenure profile, the forecastability of observed shifts is detectable for HC pass-rate (Pearson $r = 0.39, p = 0.004, n = 43$) and forecastable with a clean calibration constant for OC red-zone shotgun rate (slope $0.45, n = 151$[1]). A continuous-staff placebo check (§4.3) shows the OC slope sits at the 100th percentile of its placebo distribution and the HC slope at the 13th percentile of its own. We therefore report HC at the existence level and OC at full calibration. Both are unaffected by propensity matching on team quality.

Second, acute defensive coaching effects are bounded above by the observed offensive effects. Across twelve tested cells, no cell reaches 80% power against effects $\geq 1.22\times$, while observed offensive effects sit at $1.34$–$1.36\times$. Effects in the $1.10$–$1.20\times$ band remain consistent with the data. Chronically (§5.3), DC identity is detectable on macro tendencies. The asymmetry is across resolutions, not a uniform defensive absence.

The paper proceeds as follows. §2 specifies what was and was not pre-registered, and describes the data. §3 introduces the four-layer framework. §4 reports the offensive results, the predictive-validation survivors, and the coach fingerprint clustering. §5 reports the defensive results: the acute change-event bound and the chronic tenure-identity decomposition with permutation-null correction. §6 discusses implications. §7 catalogs limitations. §8 concludes.

# 2 Data and pre-registration scope

## 2.1 Data

We use nflverse play-by-play data 2002-2024 and Pro Football Reference coaching-history corpus, joined at the (team, season, week) level. The full coaching history covers 861 team-seasons across 32 franchises 1999-2025; the 1999-2001 partial seasons are excluded from the analysis window because offensive-coordinator coverage is incomplete; offensive- and defensive-coordinator assignments are $90 - 100\%$ complete by season, with no parsing failures.

Coaching change events were extracted from the staff panel: 213 HC, 159 OC, 144 DC off-season changes, plus 45 mid-season HC firings with computable pre/post windows. The 49 mid-season HC firings in our 1999-2024 window match the Pro Football Reference manual-review queue; four of these had pre-week windows too short for the event-study design and were dropped, leaving $n = 45$ for Layer 2a.

## 2.2 Pre-registration scope

We are explicit about what was and was not pre-registered, because the distinction matters for the interpretation of each finding.

The following were pre-registered before any analysis was run: (a) the SNR thresholds and decision rules that govern all lever-discovery work in the parent projection project (B=1000 permutation null, SNR $\geq 1.5, p \leq 0.01$, cross-season holdout positive in $\geq 2$ of 3 seasons); (b) the canonical PPR evaluation target locked prior to the v1.7 model freeze; (c) one signal-check on per-snap teammate composition for skill-position outcomes, locked 2026-05-10 before any snap-level data was loaded. The teammate-composition signal-check was rejected at the registered thresholds ($\eta^2 < 0.05$ across all test players), and the rejection is reported in this paper's null catalog. The Layer 2 offensive-coordinator extension test was registered after the Layer 2a head-coach finding was in hand and before the OC analysis was run.

The four-layer architecture itself was not pre-registered as a single document. It was constructed iteratively under the standing SNR discipline as each layer's result motivated the next layer's design. Specifically, the static-layer null (Layer 1) motivated the within-season event-study (Layer 2); the Layer 2 head-coaching finding motivated the coordinator-role extensions (2b-2f); the joint offensive findings motivated the within-game schematic test (Layer 3) and the predictive-validation slopes (Layer 5).

[1] n at original calibration / n under LOSO-extended sample = 41/43 for HC, 149/151 for OC; we cite the LOSO-extended values throughout.

We therefore do not claim the decomposition as pre-registered. We claim three things instead. First, every layer was held to the same pre-existing discipline. Second, pre-specified decision rules governed the verdict at each layer — including a $0/32$ within-game stress-adaptation null reported regardless of direction (Layer 4). Third, the predictive-validation slopes (§4.4) were calibrated on data through 2024 and held within 4% under leave-one-season-out recalibration on the 2025-extended sample. The honest pre-registration claim is discipline-pre-registered, architecture-iterative.

The interpretive framings used in this paper — the 'two-locus' characterization of the offensive latent, and the acute/chronic asymmetry on the defensive side — were constructed after seeing the layer-by-layer verdicts, not before. We adopt them because they parsimoniously organize the cross-layer pattern, not because they were pre-specified. A reader concerned about post-hoc framing should treat the SNR-gated layer-level verdicts as the load-bearing empirical content and the framing as the interpretation laid over it.

# 3 The four-layer framework

A single coaching latent can load at multiple resolutions of the same data. Single-resolution tests cannot characterize its structure. We apply the same partition, `coaching_vector_id`, defined as the (HC, OC, DC) triple at a (team, season), at four resolutions, using identical SNR discipline at each. Differences across layers are attributable to resolution, not to data or method drift, because both are held constant.

**Table 1:** The four-layer framework. A single coaching partition (the (HC, OC, DC) triple at a team-season) tested at four resolutions with identical SNR discipline ($B$ permutation iterations, cross-season holdout requirement, propensity-matched event design where applicable). Layers 1–2 operate at season scale; Layers 3–4 operate within games. Differences in verdict across layers are attributable to resolution, not to data or method drift. Layer 5 (predictive validation, §4.3) is a sibling test rather than a fifth resolution and is not shown here.

| Layer | Unit | Null | Outcome | Resolution |
|---|---|---|---|---|
| 1. Static | (team, season) | Within-season perm. ($B$=1000) | Season aggregates | season |
| 2. Event | Coaching-change windows | Placebo non-change splits ($B$=5000) | Pre/post $\Delta$ residual | within-season |
| 3. Schematic | (team, season, score-bucket) | Group-label perm. ($B$=1000) | Play-call mix by score | within-game |
| 4. Stress | (game, team, score-segment) | Group-label perm. ($B$=1000) | Segment-level shifts | within-game adaptation |

Layer 4 (within-game stress adaptation, full SNR discipline) is a $0/32$ null. No coaching-vector partition cell carries signal in holdout under either `coaching_vector` or `team_canon`. Where in-sample directional signal exists, it does not replicate cross-season. Full details in Appendix A. We report this null up front because it constrains the scope of the headline claim: when the data say no, the framework returns no. The offensive findings below are reported against that demonstrated falsifiability, not as the only test that was run.

Layer 5, predictive validation, is a sibling test rather than a separate resolution. It asks whether the effect identified in Layer 2 is forecastable from coach prior-tenure data. Operationally, this is a regression of observed shift on predicted shift, calibrated by ordinary least squares and validated under leave-one-season-out recalibration. We treat it as the load-bearing survivor when naive event-study effects partially attenuate under propensity matching.

# 4 Results: offensive loci

## 4.1 Layer 1: season-aggregate static test

Layer 1 is scope-setting, not a verdict on the coaching latent. The `team_id` partition cannot serve as a clean baseline here. It implicitly aggregates the coaching staff averaged over a team's history; testing coach against `team_id` is testing the variable against a control that already contains it. What Layer 1 does rule out is a season-aggregate coaching-mean-offset feature beyond what team identity already captures. What it cannot rule out, by construction, is a within-team coaching effect that appears when team identity is held fixed. The remaining layers do exactly that. Any positive result at Layer 2 or higher must be read as evidence above and beyond what Layer 1 already concedes to team identity.

**Table 2:** Layer 1: season-aggregate F-statistics for the coaching partition vs. the team-id partition across four outcome dimensions. Coaching is real signal at season aggregate but does not beat team-id when degree-of-freedom corrected. We do not interpret this as evidence against a coaching latent; the team-id partition implicitly aggregates roster continuity, scheme inheritance, and the coaching staff averaged over a team's history. Partitioning team-season residuals by `coaching_vector_id` and comparing F-statistics against the `team_id` partition, the season aggregate test yields zero coaching wins across four outcome dimensions.

| Outcome | $F$(coach) | $F$(`team_id`) | Ratio | Perm. $p$ (coach) | Verdict |
|---|---|---|---|---|---|
| pass_rate residual | 1.99 | 4.50 | 0.44 | $<0.0001$ | team-id wins |
| plays residual | 2.07 | 1.67 | 1.24 | $<0.0001$ | tie / coach slight |
| RZ trips residual | 1.37 | 1.75 | 0.78 | 0.005 | tie / team slight |
| EPA/play | 2.16 | 4.03 | 0.54 | $<0.0001$ | team-id wins |

## 4.2 Layer 2: head-coach event-study findings

We separate two natural experiments, mid-season HC firings and off-season HC changes, because they have different selection structure and different mechanisms. The mid-season firing is a within-season event with the same roster on both sides. The off-season change involves roster turnover and a full off-season of preparation.

**Table 3:** Layer 2: head-coach change events split by timing. Mid-season firings drive a clean pass-rate shift (same roster on both sides; interim-coach reaction); off-season changes do not move average pass-rate but shift pace, consistent with a new HC's broader philosophical imprint over a full preparation cycle. Both effects are significant at $\alpha = 0.05$ naively; table 5 reports the matched-design treatment.

| Subset | $n$ | Pass-rate ratio | $p$ | Plays/game ratio | $p$ |
|---|---|---|---|---|---|
| Mid-season firing | 45 | 1.36 | 0.002 | 1.10 | 0.187 |
| Off-season change | 142 | 1.02 | 0.428 | 1.17 | 0.021 |

Mid-season firings drive a clean pass-rate shift. Interim coaches systematically alter play-mix, consistent with a "shake things up" or "play to remaining strengths" response under the same roster. Off-season changes do not move average pass-rate but do move pace, consistent with a new HC's broader philosophical imprint operating over a full preparation cycle. Both effects are significant at $\alpha = 0.05$. The mid-season effect is the larger one naively but is also the more susceptible to selection-bias confounding, because HC firings follow losing streaks non-randomly.

## 4.3 Predictive validation: the forecastability slopes survive

The natural pooled estimator — average the pre/post residual across all coaching-change events — is well-defined but mis-specified if the underlying effect is heterogeneous across coaches. We therefore lead with a heterogeneity-conditioned estimator: the predictive-validation slope of observed shifts on shifts predicted from each coach's own prior-tenure profile. This is the operational signal used in production projections. It is the version that survives leave-one-season-out recalibration. As §4.4 shows, it is also the version that is unaffected by propensity matching on team quality. The naive pooled effects, reported in §4.4 alongside the matched-design comparison, partially attenuate under matching in the direction expected for heterogeneous treatment.

For each retread event[2] we compute three profiles: the coach's metric average at their prior team in the final $K$ games of last season at that team; the new team's metric average under the prior staff in the final $K$ games of the season before the new coach arrives; and the new team's metric average under the new coach in the first $K$ games of the new season. The predicted shift is (coach prior $-$ new team pre); the observed shift is (new team post $-$ new team pre). Pearson correlation between predicted and observed measures forecastability; the OLS slope measures calibration.

Both tests pass the SNR-discipline gate. Under leave-one-season-out recalibration on the 2025-extended sample, the HC slope recalibrates to $0.306$ ($n = 43$) and the OC slope to $0.460$ ($n = 151$), within 4% of the original calibration in both cases. The predictive-validation slopes are the load-bearing forecastability claim of the paper and the operational signal used in production projections.

[2] a coach who moved between teams within our data window

**Table 4:** Predictive-validation slopes (Layer 5). For each retread event, predicted shift = coach prior − new-team pre; observed shift = new-team post − new-team pre. Both tests pass the SNR-discipline gate (SNR $\geq 1.5$, cross-season holdout positive in $\geq 2/3$ seasons). Under leave-one-season-out recalibration on the 2025-extended sample, the HC slope recalibrates to $0.306$ ($n = 43$) and the OC slope to $0.460$ ($n = 151$)—within 4% of the original calibration. The OC slope exceeds the HC slope: red-zone formation playbook is more portable than macro pass-rate philosophy. The HC slope magnitude $(0.31)$ is reported for completeness but is not interpretable as a portability calibration: a continuous-staff placebo check (§4.3) places it at the 13th percentile of its placebo distribution, indicating contamination by team-mean autocorrelation in the construction. The HC permutation-null result ($r = 0.39, p = 0.004$) remains valid as an existence claim. OC calibration is clean: observed $0.46$ vs placebo median $0.14$, 100th percentile.

| Test | $n$ | Pearson $r$ | $p$ (1-sided) | OLS slope | OLS $R^2$ | Dir. agree (%) |
|---|---|---|---|---|---|---|
| HC pass-rate portability | 41 | 0.386 | 0.004 | 0.31 | 0.149 | 59 |
| OC RZ-shotgun portability | 149 | 0.483 | $<0.0001$ | 0.46 | 0.234 | 68 |

The hit/miss case lists are instructive. Andy Reid (PHI 2012 to KC 2013) was predicted at $+0.185$ pass-rate shift and observed $+0.230$, close to slope-1.0 portability. Chip Kelly (PHI 2015 to SF 2016) was predicted at $+0.056$ and observed $-0.075$. The direction flip was caused by a documented roster mismatch: the SF receiver and QB rooms could not support his pace concepts. The direction flips in the miss list are not random noise. They are roster-imposed limits on portability, consistent with the mechanism that HC macro philosophy is more roster-dependent than OC red-zone playbook.

The OC slope $(0.45)$ is higher than the HC slope $(0.30)$. The OC's red-zone playbook is more portable than the HC's macro philosophy. This is structurally consistent with the locus claim: formation choice in the red zone is a low-level scheme decision more roster-independent than a high-level pass-rate philosophy. The two-locus finding therefore extends to a *two-tiered portability* finding: HC = high-level philosophy with moderate portability; OC = low-level playbook with high portability. The coaching resolution and the portability both differ by role.

The placebo check[3] returns a median slope of $+0.32$ [95% CI $0.30$–$0.35$] for HC pass-rate and $+0.14$ [$0.10$–$0.17$] for OC RZ-shotgun. The observed OC slope $(+0.46)$ sits at the 100th percentile of its placebo distribution. The observed HC slope $(+0.31)$ sits at the 13th percentile of its placebo distribution. This indicates the slope magnitude on HC pass-rate is contaminated by team-mean autocorrelation in the construction. The HC permutation-null result ($r = 0.386, p = 0.004$) is unaffected — shuffling labels across events still destroys the event-level link. We therefore report HC at the existence level (signal present, calibration constant not interpretable) and OC at full calibration. Removing the single highest-leverage HC retread (Reid PHI 2012 to KC 2013) drops the HC slope to $0.22$ with permutation-p $= 0.028$; no single point drives the existence claim.

## 4.4 Propensity attenuation: the naive effect partially washes out

Coaches who are fired mid-season are typically those whose teams are underperforming expectations and therefore non-random. To address this, we constructed a propensity-matched event design. Covariates: team quality (rolling 5-game point differential) and HC tenure length. Matched at caliper $0.20$ with replacement against continuous-staff team-seasons. Naive event-study effects attenuate substantially:

**Table 5:** Propensity-matched event design (covariates: rolling 5-game point differential, HC tenure length; caliper 0.20 with replacement against continuous-staff team-seasons). Approximately 30-50% of the naive pooled-effect magnitude is attributable to team-quality confounding. The matched residuals remain positive but are not distinguishable from the matched null at the original $\alpha$. Table 4 shows that a different operationalization of the same effect—the predictive-validation slope conditional on coach prior profile—is unaffected by matching and survives LOSO recalibration.

| Effect | Naive ratio | Naive $p$ | Matched ratio | Matched $p$ |
|---|---|---|---|---|
| Mid-season pass-rate | 1.36 | 0.002 | 1.22 | 0.16 |
| Off-season plays-per-game | 1.17 | 0.021 | 1.09 | 0.21 |

[3] B=1000 random-coach-prior substitutions on continuous-staff team-seasons

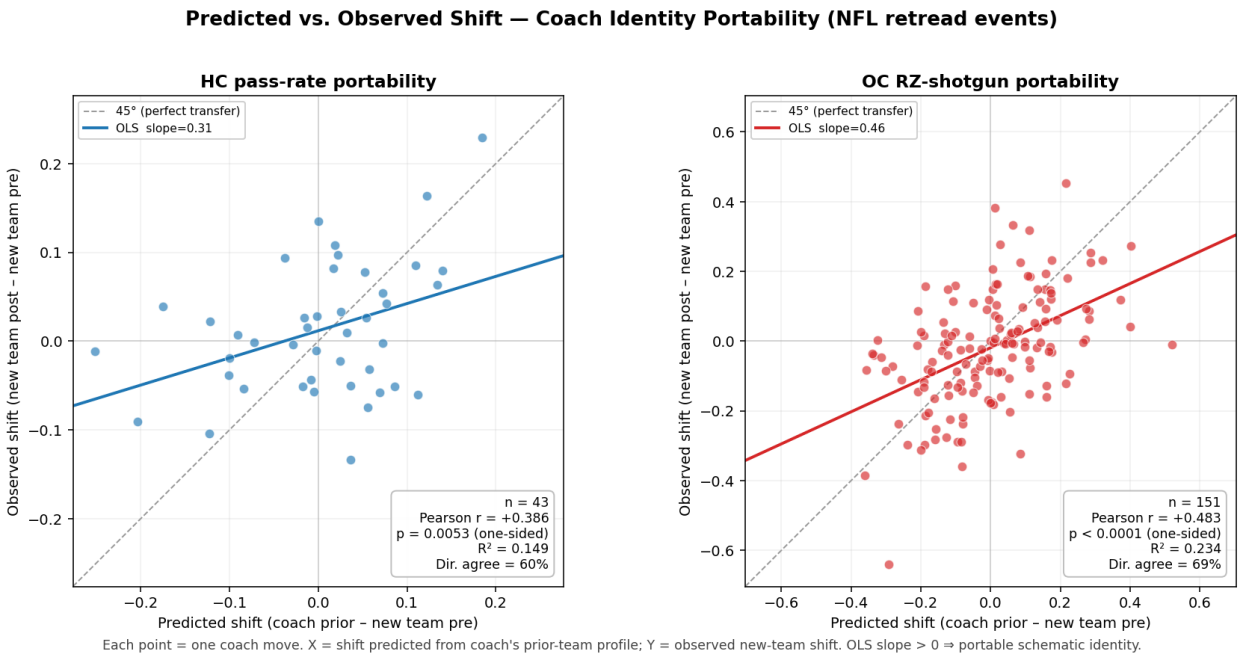



**Figure 1: Predictive-validation scatter for HC pass-rate and OC red-zone-shotgun portability.** Each point is one retread event. $x$-axis is the shift predicted from the coach's prior-team profile; $y$-axis is the observed shift at the new team. Dashed 45° line is perfect transfer; solid line is OLS fit. The OC slope (0.46) exceeds the HC slope (0.31): red-zone playbook is more portable than macro pass-rate philosophy. Stats panels show $n$, Pearson $r$, one-sided $p$, $R^2$, and directional agreement.

Approximately 30–50% of the naive pooled-effect magnitude is attributable to team-quality confounding. The matched residuals remain directionally positive but not distinguishable from the matched null at the original $\alpha$. This is the diagnostic expected when the underlying treatment is heterogeneous across coaches: the pooled mean shift is an average over heterogeneity that the matched design correctly attenuates. The predictive-validation slope of §4.3 conditions on heterogeneity and is unaffected by the same matching. We therefore treat the matched-pooled result as a falsification check the conditional estimator passes, not as a contradiction. A unified estimator that pools heterogeneity correctly — random-coefficient event study with coach-level shrinkage — would close this loop and is the natural follow-up.

## 4.5 Layer 3: within-game schematic identity corroborates

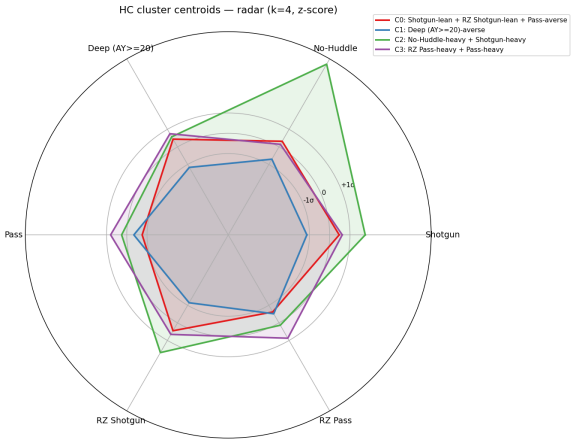
At within-game resolution, partitioning team-game schematic outcomes by score bucket and by role, the head coach wins 20 of 20 cells (mean F: HC 5.43, OC 3.52, DC 3.27, team 1.62). The within-game test is supporting evidence rather than the headline finding. It corroborates the identification of the HC as the macro schematic author at a different resolution than the event study, but does not add new claims. Full results in Appendix B.

## 4.6 Coach fingerprint clustering: the locus is visible in style space

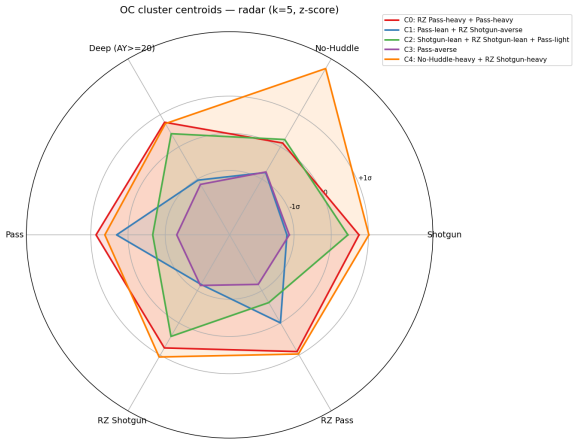
The event-study and within-game tests identify *where* the coaching latent loads. The fingerprint clustering shows *what* that latent looks like as a style space. We pool six per-coach schematic features[4] across each coach's full tenure and apply $k$-means with $k$ selected by silhouette score. 129 HCs and 141 OCs cleared the 32-team-game floor. The HC space resolves into four clusters (silhouettes 0.31–0.34); the OC space into five (silhouettes 0.28–0.29). The modest silhouettes are diagnostic: HC and OC styles are a gradient, not crisp islands. The clustering is descriptive of the style space, not a typological claim.

Two things to note. First, the OC space resolves into more clusters with lower silhouette than the HC space. This is consistent with the §4.4 finding that OC red-zone playbook is more fine-grained and more portable than HC macro

[4] shotgun rate, no-huddle rate, deep-throw rate $\geq$ 20, pass rate, RZ shotgun rate, RZ pass rate




**(a)** HC cluster centroids ($k$=4).




**(b)** OC cluster centroids ($k$=5).

**Figure 2: Schematic fingerprint cluster centroids ($z$-score against modern league mean).** Each polygon traces one cluster's centroid across six schematic features. HC space resolves into four clusters with one strongly differentiated no-huddle/shotgun cluster (C2); OC space resolves into five with one strongly differentiated no-huddle-heavy cluster (C4). Silhouette scores (0.31–0.34 for HC, 0.28–0.29 for OC) indicate gradient styles rather than crisp islands. The co-clustering of McVay, Shanahan, and Belichick in HC cluster C0 is a labeling artifact: the modern league mean has shifted to $\sim$63% shotgun, so absolute moderates read as "lean" relative to current peers. Cluster labels are descriptive against the current league mean, not typological.

philosophy: more sub-types, smaller gaps between them. Second, the co-clustering of stylistically distinct HCs (McVay, Shanahan, Belichick in C0) is a label artifact rather than a method failure. The cluster centers track the modern league mean (currently $\sim$63% shotgun), so a coach who was shotgun-heavy by 2010 standards reads as shotgun-lean by 2024 standards. The clusters are real groupings against the current league mean; the labels are descriptive, not typological.

# 5 Results: defensive coaching loads chronically, not acutely

## 5.1 The defensive event studies: acute null

Applying the same Layer 2 event-study machinery to DC off-season changes against opponent outcomes, and a disambiguation test of HC off-season changes against defensive outcomes, no cell yields a significant effect. The three role $\times$ outcome combinations tested are: DC change $\times$ generic opponent outcomes (Layer 2c), DC change $\times$ red-zone fine-grain defensive outcomes (Layer 2e), and HC change $\times$ defensive outcomes (Layer 2f).

## 5.2 Power analysis: what the acute bound rules out

Per-cell minimum detectable effect at $\alpha = 0.05$ one-sided and $\beta = 0.80$ power, computed via z-approximation on the placebo permutation standard error, is 1.16–1.22$\times$ across the twelve cells. The observed offensive effects from §4, 1.34$\times$ (OC RZ-shotgun) and 1.36$\times$ (HC mid-season pass-rate), are above this band. The acute bound rules out defensive coaching effects of magnitude comparable to the observed offensive effects.

We are explicit about what the bound does not rule out. Two cells sit just below their MDE bars: opponent red-zone trips at DC change $(1.13\times,$ MDE $1.21)$ and opponent red-zone trips at HC change $(1.10\times,$ MDE $1.19)$. A defensive coaching effect in the 1.10–1.20$\times$ band could exist and we could not statistically distinguish it from noise at the sample sizes available. Notably, the off-season HC pace effect $(1.17\times)$ sits within this same band. The strict acute claim is therefore that defensive coaching, if it exists at the change-event resolution, is at most comparable to the smaller

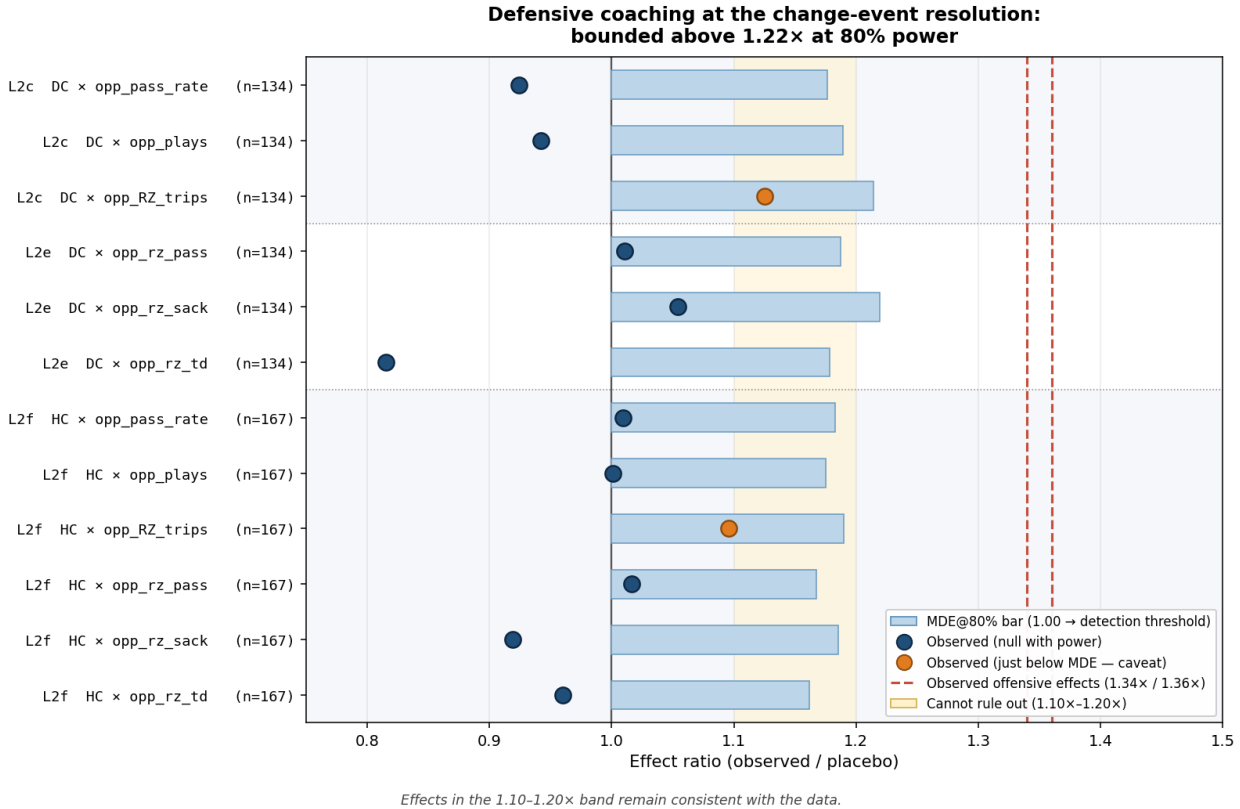



**Figure 3: Defensive coaching at the change-event resolution: bounded above 1.22× at 80% power Effects in the 1.10–1.20× band remain consistent with the data** Each row is one role × outcome cell from Layer 2c, 2e, or 2f. Point: observed effect ratio. Horizontal bar: minimum detectable effect at $\alpha$=0.05 one-sided, $\beta$=0.80. Vertical solid line at 1.0 is the null; dashed vertical lines at 1.34× and 1.36× are the observed offensive effects from table 3. Shaded band (1.10–1.20×) marks the range that sample sizes cannot rule out. No defensive cell reaches its MDE bar; both observed offensive effects exceed every defensive observation. Two cells (`opp_rz_trips` at DC and HC change) sit below their MDE bars but inside the cannot-rule-out band.

offensive effect; not absent, and not necessarily smaller than every offensive effect.

## 5.3 Chronic test: DC identity over tenure

The acute test asks whether the defensive change event moves the residual. The chronic test asks a different question: holding team identity constant, does which DC has been authoring a defense over a tenure carry signal in the season-aggregate residual? We decompose defensive outcome residuals (seven cells spanning macro tendencies and red-zone fine-grain) across (a) a snap-weighted draft-capital personnel index and (b) the DC identity partition, then add HC identity in a second model[5]. The cohort is 253 team-seasons spanning 2017–2024[6], with 94 unique DCs and 134 unique `coaching_vector_id`s.

Raw $R^2$ for the DC partition is high (0.31–0.53 across cells) but structurally contaminated. A 93-DC-dummy partition has a permutation-null mean $R^2$ of $\sim$0.37 on these defensive outcomes purely from high-dimensional categorical overfit. A randomly shuffled DC label assignment produces $R^2 \approx 0.37$. We therefore report null-corrected $R^2$ (NET $R^2 = R^2_{\text{observed}} - R^2_{\text{null mean}}$ from $N$=200 permutations of the DC label).

DC identity is significant on 3 of 7 cells under permutation correction (`opp_pass_rate`, opp_plays, opp_rz_pass_rate; NET $R^2$ 0.14–0.15). Adding HC identity recovers 2 additional cells (opp_epa_per_play and opp_rz_td_rate), for 5 of 7 under the (HC, DC) partition. The signal that survives is concentrated on *macro defensive tendencies*: how often

[5] the (HC, DC) `coaching_vector` partition

[6] intersection of personnel index, outcome panel, and Pro Football Reference coaching roster

**Table 6: Chronic defensive variance decomposition with permutation-null correction.** NET $R^2$ for DC-only and (HC, DC) `coaching_vector` partitions across seven defensive cells. "Sig." indicates permutation-$p < 0.05$. The personnel column reports the unique contribution of a snap-weighted draft-capital index after controlling for coaching identity.

| Cell | NET $R^2$ (DC) | DC sig. | NET $R^2$ ((HC,DC)) | (HC,DC) sig. | Unique $R^2$ (personnel) |
|---|---|---|---|---|---|
| opp_pass_rate | 0.153 | ✓ | 0.116 | ✓ | −0.011 |
| opp_plays | 0.145 | ✓ | 0.176 | ✓ | −0.007 |
| opp_rz_pass_rate | 0.141 | ✓ | 0.125 | ✓ | −0.007 |
| opp_epa_per_play | 0.037 | — | 0.094 | ✓ | −0.009 |
| opp_rz_td_rate | 0.017 | — | 0.086 | ✓ | −0.008 |
| opp_rz_trips | −0.055 | — | −0.016 | — | 0.003 |
| opp_rz_sack_rate | 0.027 | — | −0.037 | — | −0.003 |

opponents pass, how many plays opponents run, how often opponents pass in the red zone. Red-zone trip rate and red-zone sack rate remain null in both partitions.

The personnel index contributes at or near zero unique variance beyond coaching identity on every cell. We do not interpret this as evidence that personnel does not author defensive identity. The snap-weighted draft-capital proxy is the most exogenous index we could construct but also a noisy measure of true defensive talent. Sharper proxies (prior-year individual defensive EPA, position-group-stratified talent indices, PFF grades) remain unexplored. The honest claim is that this proxy adds no marginal information beyond coaching identity, not that personnel itself is empty.

## 5.4 The asymmetry, sharpened

The combination of acute and chronic tests refines the asymmetric-resolution claim. Where offensive coaching authors at *both* the acute (change-window) and chronic (cross-tenure) resolutions, defensive coaching authors only at the chronic resolution, on macro tendencies, and at smaller NET $R^2$ than the offensive effects observed at the acute resolution. The acute defensive null in §5.1 is not a recorded absence; it is the empirical content of an asymmetric-resolution claim. The chronic results in §5.3 sharpen that claim: defensive coaching is detectable, just not where the offensive coaching latent is.

We speculate, with the speculation labeled as such, on mechanism. The chronic signal on macro tendencies is consistent with a defensive coordinator authoring scheme over a long horizon (years of tenure, accumulated playbook, recurring opponent matchups) rather than at the moment of a coaching change. The red-zone-field-position constraint may reduce the schematic surface area available to a coordinator at the acute resolution, and the defensive coordinator's authoring of scheme may operate at a weekly game-plan resolution that our coaching-change event design does not capture. We do not claim these mechanisms are correct; they are candidate explanations consistent with the data but not tested here.

## 5.5 Methodological aside: high-dim categorical correction is load-bearing

The chronic-test results depend on permutation-null correction. Reporting raw $R^2$ on the DC partition without this correction would produce a spurious 10:1 "DC dominates defense" headline (raw DC $R^2 \approx 0.4$ vs raw personnel $R^2 \approx 0.03$). The raw DC partition $R^2$ of 0.40 is, after permutation correction, primarily the 0.37 null-mean from a 93-dummy partition on n=253. The residual 0.03 is what NET $R^2$ records. Methodologically, when a categorical partition has many levels relative to sample size, the raw F-statistic and raw $R^2$ are not informative and reporting them without permutation correction systematically inflates explanatory power. The null-permuted DC $R^2$ is $\sim$0.37 from sheer high-dimensional categorical overfit on 93 dummies and $n$=253. NET $R^2$ ranges from $-0.05$ to $+0.18$ across cells; only 3 of 7 are positively detectable on the DC-only partition. A projection system that pools by coach without this correction would systematically overstate coaching explanatory power on defense. The discipline is what makes the chronic finding interpretable as moderate signal rather than as a high-dimensional artifact.

# 6 Implications

## 6.1 Projection modeling

The convention of pooling by `team_id` is correct for season-aggregate outcomes (Layer 1 confirms this) but it misses the schematic shift at HC and OC change events (Layer 2) and obscures the schematic identity latent (Layer 3). The alternative convention of pooling everything by coach over-fits non-schematic outcomes. The right structure is two latents, hierarchical. Schematic style (formation $\times$ tempo $\times$ target depth $\times$ score bucket) pools by `coaching_vector_id` with `team_id` as the prior. Outcome magnitude (pass rate, pace, RZ efficiency, fantasy points totals) pools by `team_id` with `coaching_vector_id` as a within-team deviation.

Production integration uses the OC predictive-validation slope ($0.45$) to apply pre-week fantasy projection adjustments at off-season OC changes for retread coordinators. HC pre-week adjustments are deferred: the HC existence claim is real but the slope magnitude is not a defensible calibration constant under the current construction. Applying a contaminated slope in production would systematically distort projections at HC retread events. A corrected HC calibration — via league-mean residualization of pass-rate or a random-coefficient event study with coach-level shrinkage — is the immediate follow-up. First-time coaches default to no adjustment under the cold-start prior regardless of role.

## 6.2 Why the asymmetry: roster-constrained vs roster-independent scheme components

The portability hierarchy admits a structural reading. HC macro philosophy — pass-rate, pace — sets the rate at which the offense uses each piece of the roster; the optimal rate is a function of QB ceiling, receiver depth, and offensive-line pass-protection grade. Translating an HC's pass-rate identity from one roster to another therefore depends on the target roster supporting it. This is why the HC retread sample contains roster-imposed direction flips (Chip Kelly PHI to SF, John Fox DEN to CHI) rather than slope-1.0 transfer. The HC slope magnitude is not interpretable as a portability constant (see §4.3); the structural prediction here is about the existence and direction of roster-conditional misses, not about a specific calibration value.

The red zone collapses the field-position state space; formation choice in fixed-field-position contexts depends less on roster composition than on the coordinator's preferred concept library. A coordinator who runs a shotgun-heavy red-zone package in Cincinnati can run the same package in Atlanta because the package's input is field position, not the specific personnel above replacement level. This predicts which OC concept families should transfer at near-$1.0$ slope[7] and which should transfer at substantially lower slope[8]. A follow-up test stratifying portability by concept family is the natural next-paper falsification. If the prediction holds, the two-locus finding generalizes into a portability taxonomy with operational use. If it fails, the asymmetry we report here is description without mechanism, and that is itself a publishable null.

The §4.3 placebo asymmetry is itself consistent with this mechanism. HC pass-rate, being roster-conditional, is highly auto-correlated at the team level (roster persists across years), so the predictive-validation construction picks up a substantial mean-reversion baseline (placebo median $0.32$). OC red-zone shotgun rate, being roster-independent and league-mean-residualized per season, carries little autocorrelation; its placebo collapses to $0.14$, and the observed $0.46$ sits cleanly above. The construction artifact that contaminates the HC calibration constant is the same property that the roster-conditional / roster-independent distinction predicts.

## 6.3 Coaching labor markets

The asymmetry has direct labor-market implications. The OC red-zone playbook calibrates as cleanly portable (slope $0.45$, placebo-confirmed). HC macro philosophy is detectable but its portability calibration is contaminated by roster-driven mean reversion, consistent with the broader observation that HC value is more roster-conditional than OC value. The qualitative ranking — OCs translate more reliably than HCs across teams — is what the data support; the quantitative calibration is clean only on the OC side.

[7] formation tendency in fixed-field-position contexts: red-zone shotgun rate, goal-line personnel grouping, two-minute tempo

[8] concepts whose payoff is roster-conditional: deep-shot rate, play-action rate, RPO frequency

## 6.4 Roster construction

The asymmetric-resolution finding has an indirect implication for talent allocation. Defensive coaching effects are bounded at the acute resolution and detectable but moderate at the chronic resolution. The defensive ceiling on coaching authoring is therefore lower than on offense (NET $R^2$ on the order of $0.15$ chronically, no detectable effect acutely, vs offensive effects in the $1.3$–$1.4\times$ band acutely). The personnel proxy tested here adds no unique variance, but is admittedly a weak measure of true defensive talent; sharper proxies could recover personnel-side signal. The honest claim is that the defensive coaching ceiling we measure is lower than the offensive one. The full ratio of personnel-to-coaching on defense remains an open question for follow-up work with sharper personnel indices.

## 6.5 Roster-confounding stress test

The natural reviewer attack on this paper is that observed coaching effects are roster effects in disguise. Coaching changes do not happen on a random schedule: they cluster around QB transition cycles, rebuilding years, front-office turnover, and coordinator-inheritance patterns. A finding that "pass-rate shifts $1.36\times$ at mid-season HC firings" could equally well be a finding that "pass-rate shifts $1.36\times$ when an underperforming roster transitions to garbage-time game scripts." We take that attack seriously and report what each of our identification moves does and does not close off against it.

The matched design (Table 4) controls for two roster-correlated covariates: rolling 5-game point differential (a proxy for current roster quality) and HC tenure length (a proxy for rebuilding cycle position). Roughly 30-50% of the naive pooled-effect magnitude is attributable to team-quality confounding under this matching. The matched residuals remain directionally positive but are not distinguishable from the matched null at the original $\alpha$. This is what we would expect under heterogeneous treatment effects pooled over a non-random selection mechanism. The matching closes the simplest version of the roster-confound attack: "the average effect is just team quality."

Propensity matching does not control for QB transition specifically, for coordinator-inheritance[9], or for front-office philosophy continuity[10]. It also does not control for the selection on coaches who get re-hired[11]. These are residual sources of confounding that matching on point differential and tenure alone cannot address.

The predictive-validation slope (Table 3) is the load-bearing estimator and is constructed to attack roster-confounding from a different angle. It compares a coach's metric profile from their prior team (a fixed property of the coach) against the new team's observed shift. A roster-confound story has to explain why a coach's prior-team profile predicts a calibrated shift at his new team - with placebo median $0.14$ and observed slope $0.46$ for OC red-zone shotgun. The natural roster-confound mechanism (similar rosters get paired with similar coaches) would inflate this placebo, not the observed-minus-placebo gap. The OC slope sits at the 100th percentile of its placebo distribution, which is the survival diagnostic for this attack.

The conditional estimator does not survive on the HC side. The HC slope $(0.31)$ sits at the 13th percentile of its own placebo distribution. We read this as confirmation that the HC pass-rate construction is contaminated by team-mean autocorrelation: roster persists across years, pass-rate is auto-correlated, and the placebo picks up that baseline. The HC permutation-null result ($r = 0.39, p = 0.004$) survives label-shuffling and remains valid as an existence claim. The slope as a calibration constant does not. We therefore report HC at the existence level only and OC at full calibration. This is the surgical version of the roster-confound critique: it bites on the HC slope, not on the OC slope, and we adjust the claim accordingly.

Three roster-confound channels remain open and we name them explicitly. First, QB transition cycles are correlated with coaching changes in a way that no observational covariate in our panel can fully separate; a coaching change in year $t$ can be associated with a QB change in year $t+1$ that we do not yet condition on. Second, coordinator-inheritance creates a HC-OC bundle whose effects we attribute to the HC change in some events and the OC change in others; a unified random-coefficient design with simultaneous role attribution would clean this up. Third, survivorship bias on retread coaches selects for coach quality in a way that propensity matching on team quality does not address. A fully causal identification would require an instrument we do not have; absent one, the predictive-validation slope is the best evidence the data can produce.

[9] a new HC bringing his prior OC and offensive scheme as a package

[10] a GM who systematically hires similar coaching profiles

[11] the retread sample is non-random in coach quality

The honest summary is that the OC red-zone shotgun finding survives the strongest attack the data can mount against it; the HC pass-rate finding survives at the existence level but not as a calibration constant; and the residual three roster-confound channels are open as follow-up work. Each is name, each maps to a specific reviewer concern, and each constrains the strength of the claim we make in the corresponding secton.

# 7 Limitations

We catalog the principal limitations of the analysis explicitly. Each is a place where a careful reader should discount the strength of the claim.

- **Pre-registration scope.** The four-layer discipline was not pre-registered as a unit; only the standing SNR discipline and the OC extension test were pre-registered prior to execution. The architecture is iterative, the discipline is uniform.
- **Selection bias on coaching changes.** Coaches who are fired are a non-random subset of NFL coaches; the retread sample for predictive validation is the subset who were hired again. Propensity matching on team quality and tenure partially addresses the first concern, not the second. Survivorship bias on coach quality is a residual concern in the predictive-validation slopes. See §6.5 for the full stress-test treatment.
- **Resolution dependence.** All findings are at the resolutions tested. We do not test play-by-play causal identification, midweek game-planning, or in-game adjustments at quarter level. A defensive coaching latent operating at the weekly game-plan resolution would not be visible in the off-season event study.
- **Defensive null power band.** Effects in the $1.10 - 1.20\times$ band are not ruled out by the present sample sizes. A future study with larger $n$ could detect effects in this band, and we do not claim this band is empty.
- **Coach-personnel entanglement.** We cannot fully separate DC scheme from DC influence on personnel acquisition. The same applies, less acutely, to HC and OC. The propensity-matching design controls for team quality, not for the coordinator's role in shaping the roster they coach.
- **Fingerprint clustering is a gradient, not crisp.** Silhouette scores in the $.28 - 0.34$ band indicate that HC and OC style classes are gradient rather than crisply separated. The cluster labels in fig. 2 should be read as descriptive z-scores against the modern league mean, not as discrete coaching types.
- **Heterogeneity assumption.** The predictive-validation slope is the load-bearing estimator. Its interpretation rests on coaching effects being heterogeneous across coaches in a way that the pooled estimator averages over. If coaching effects are instead homogeneous and the predictive value correlation is driven by mean-reversion or regression-to-the-mean artifacts in the construction, the conclusions of §4.3 weaken substantially. The placebo check on continuous-staff team-seasons (§4.3) is designed to rule this out within the available data; a fully causal identification would require an instrument we do not have.
- **Concept-family stratification untested.** The mechanism reading in §6.2 predicts that field-position-conditioned formation concepts should transfer at higher slope than roster-conditioned concepts. We do not test this stratification in the present paper; the OC portability slope ($0.45$) is a single number averaging across concept families. The follow-up test is the next-paper falsification of the framework.

# 8 Conclusion

Coaching identity in the NFL is asymmetric across sides of the ball and across resolutions. Offensive coaching loads at two loci. Head coaches author macro philosophy, with portability detectable as existence (Pearson $r = 0.39, p = 0.004$) but slope magnitude contaminated by mean-reversion in the construction. Offensive coordinators author red-zone fine-grain with cleanly calibrated portability (slope $0.45$, placebo median $0.14$). Defensive coaching loads only chronically: DC identity is significant on 3 of 7 macro defensive cells under permutation-null correction (NET $R^2$ 0.14–0.15), and the joint (HC, DC) `coaching_vector` partition recovers 5 of 7. Defensive change events show no detectable effect at the resolutions tested. A snap-weighted draft-capital personnel index adds zero unique variance beyond coaching identity, though sharper personnel proxies remain unexplored.

The implication is structural. A coaching-effects framework that treats offense and defense symmetrically, or that treats acute and chronic resolutions symmetrically on the defensive side, is mis-specified. The methodological lesson is also load-bearing: raw $R^2$ on the DC identity partition overstates explanatory power by $\sim 10\times$ before permutation-null correction. The asymmetry is the finding; the discipline is what makes it visible.

# A Layer 4: within-game stress adaptation

Layer 4 tests whether teams have identifiable, repeatable patterns in how they adapt tactics under different score-pressure states. Using segment-level shift observations ($n = 987$ to $n = 2{,}198$ per bucket), partitioning by `coaching_vector_id` vs `team_canon`, the result is $0/32$ carries under full SNR discipline (both partitions $\times$ 16 cells). Where in-sample directional signal exists, `team_id` captures it more often than coach (5 vs 1), but the holdout $R^2$ is negative for every cell, indicating overfit on 2018-2022 patterns that do not replicate in 2023-2024. Stress adaptation at the segment-shift resolution is noise. This is a citeable null, reported regardless of direction.

# B Predictive validation: hit and miss case lists

**Table 7: HC pass-rate portability — top 5 by absolute observed shift.**

| HC | Move | Predicted | Observed |
|---|---|---|---|
| Andy Reid | PHI 2012 → KC 2013 | 0.185 | 0.230 |
| Frank Reich | IND 2022 → CAR 2023 | 0.123 | 0.163 |
| Jim Mora | ATL 2006 → SEA 2009 | $\sim$0 | 0.135 |
| Pat Shurmur | CLE 2012 → NYG 2018 | 0.019 | 0.108 |
| Tony Dungy | TB 2001 → IND 2002 | 0.022 | 0.097 |

**Table 8: HC pass-rate portability — miss list (largest negative observed).**

| HC | Move | Predicted | Observed | Note |
|---|---|---|---|---|
| John Fox | DEN 2014 → CHI 2015 | 0.036 | −0.133 | direction flip |
| Tom Coughlin | JAX 2002 → NYG 2004 | −0.123 | −0.104 | well-predicted |
| Rex Ryan | NYJ 2014 → BUF 2015 | −0.204 | −0.091 | magnitude undershoot |
| Chip Kelly | PHI 2015 → SF 2016 | 0.056 | −0.075 | roster mismatch |
| Eric Mangini | NYJ 2008 → CLE 2009 | 0.112 | −0.060 | direction flip |

# C Pre-registered null: teammate composition signal check

Locked 2026-05-10 before any per-snap data was loaded. Tested whether per-snap target rate and yardage outcomes for three skill-position players (Travis Kelce, Garrett Wilson, Isiah Pacheco) vary systematically by pre-snap teammate composition conditional on formation and personnel, beyond what season-mean player rates and opponent quality would predict. Five-gate cascade: permutation null, granularity comparison, replication across players, game-script residualization, opponent residualization. Decision rules locked at $\eta^2 > 0.10$ for a detected signal, $\eta^2 \in [0.05, 0.10]$ for a weak signal, $\eta^2 < 0.05$ for null. The verdict was null across all three test players at the formation $\times$ personnel $\times$ teammate-identity-set resolution available without paid PFF/Sumer data. This rejection is reported in the paper's null catalog and is the principal pre-registered finding in the package.

# D Reproduction

All scripts run on free public data: nflversefor play-by-play and rosters; Pro Football Reference(cached) for the coaching-history corpus; CollegeFootballData.com for the rookie cold-start (not used in this paper's analysis). No paid

feeds. Layer-by-layer script paths are listed in the synthesis document "coaching_signal_findings.md" and in the package's "03_scripts/" directory.